Honors Geometry – Chapter 4 Test PRACTICE

Full Name: _____

Transformations • Sections 4.1 – 4.3 Period: _____ _____ / 40

Show all work in the space provided. Check your work. Write each final answer on the line. Label every image with primes.

1. Write the component form of the vector from A(4, 3) to B(−1, 5). (2)

⟨ _____ , _____ ⟩

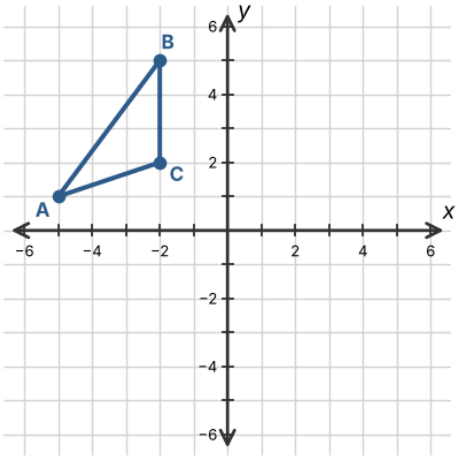
2. Write a rule for the translation that maps P(−4, 0) to P′(−10, −7). (2)

(x, y) → (_____)

3. Q′(4, 4) is the image of Q after a translation by ⟨1, −5⟩. Find Q. (2)

Q(_____ , _____)

4. Graph △ABC with A(−5, 1), B(−2, 5), C(−2, 2) and its image after a translation by ⟨7, 1⟩. (3)



5. Translate by (x, y) → (x + 5, y − 4), then by (x, y) → (x − 2, y + 5). Write ONE rule that does both. (2)

(x, y) → (_____)

6. Reflect the point (−7, 2) in each line. (4)

x-axis _____ y-axis _____

y = x _____ y = 3 _____

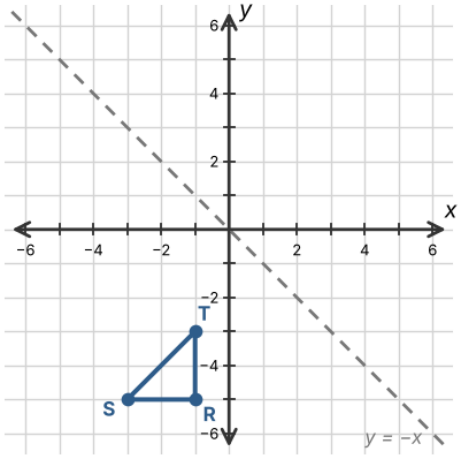
7. Reflect M(2, 3) in the line x = −1. (2)

M′(_____ , _____)

8. A reflection maps A(−6, −6) to A′(0, −6). Write the equation of the line of reflection. (2)

line: _____

9. Graph △RST with R(−1, −5), S(−3, −5), T(−1, −3) and its reflection in the line y = −x. (3)

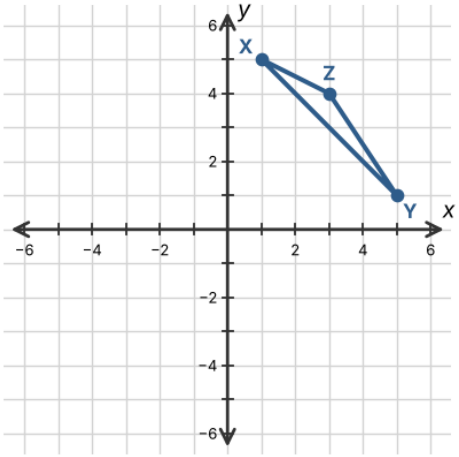


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10. Rotate the point (1, 5) about the origin by (a) 90° (b) 180° (c) 270°. (3)

(a) _____ (b) _____ (c) _____

11. Graph △XYZ with X(1, 5), Y(5, 1), Z(3, 4) and its image after a 90° rotation about the origin. (3)



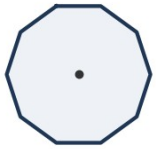
12. Glide reflection: translate by ⟨0, 1⟩, then reflect in the y-axis. Find the final image of △JKL with J(5, 5), K(4, 1), L(4, 5). (3)

J″ _____ K″ _____ L″ _____

13. Name each transformation. (a) (x, y) → (−x, y) (b) (x, y) → (y, −x) (2)

(a) _____ (b) _____

14. The figure is a regular decagon. What is the smallest angle of rotation that maps it onto itself? How many lines of symmetry does it have? (2)



angle: _____ lines: _____

15. Error analysis. Dante rotated (−4, 1) 90° counterclockwise about the origin and got (1, 4). Name the mistake, then give the correct image. (2)

correct image: _____

16. △ABC is reflected in the y-axis to form △A′B′C′. AB = 10 cm and m∠B = 42°. Find A′B′ and m∠B′. Explain in one sentence. (2)

A′B′ = _____ m∠B′ = _____

17. What is the image of (−6, −2) after a 270° rotation about the origin? (F) (2, −6) (G) (−2, 6) (H) (−2, −6) (J) (6, 2) (1)